

YUHAN JIANG

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Summary

- Double-degree MSc in Computer Science (University of Trento) and Data Science (SUPSI), EIT Manufacturing programme; research intern at Huawei Pisa on multimodal agents and LLM post-training.
- Two first-author journal papers accepted in 2026: IEEE Transactions on Geoscience and Remote Sensing and TMLR.
- GeoSelect sets the training-free state of the art on RRSIS-D and RISBench; DeliverMem (Huawei) sets the state of the art on DMV-Bench and leads published memory-augmented agents on MemLens.
- Research focus: multimodal agents, LLM post-training, embodied AI and mechanistic interpretability, with several projects run independently from problem definition to paper.

Education

EIT Manufacturing Master & Doctoral School (Double Degree)

Data Science and AI for a Competitive Manufacturing · Scholarship: full tuition waiver + 10,000 EUR/yr

- **Università di Trento (UNITN), Italy** Sep 2023 - Mar 2026
MSc, Computer Science · GPA 98/110 · Degree awarded Mar 2026
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI)** Sep 2023 - Dec 2026
Master of Engineering (MSE), Data Science · GPA 5.1/6.0 · Thesis defended Mar 2026, diploma expected Dec 2026

Anhui University (Project 211), China

Sep 2019 - Jun 2023

BEng, Intelligent Science and Technology · Average 84.34/100

Scholarships: Academic Science and Technology Scholarship, First Prize (2021); Outstanding Academic Performance, Second Prize (2022)

Research Experience

Huawei Pisa Research Center

Apr 2026 - Oct 2026

Research Intern

Pisa, Italy

Independent research: memory for multimodal agents (first-author paper under review)

- **Controlled experiments located the bottleneck at the delivery step.** Holding the retrieved set fixed and changing one variable at a time, delivering the original images is worth +13.87 points and perfect retrieval only +2.31 on Qwen3-VL-8B; the delivery gain grows on a stronger backbone (+17.92 on Qwen3-VL-32B).
- **Proposed DeliverMem, a training-free delivery method that leaves the memory store unchanged** (original modality, readable identifiers and temporal position, plus a retrieval-side adaptation): **state of the art on DMV-Bench**, ahead of the best prior method at every setting on both backbones by 7.63 to 22.62 points; on MemLens it leads the strongest published memory-augmented agent by at least 8.20 points at all four context lengths, on a tenth to a seventieth of the full-context input. At MemLens 32K with dev-fitted selection it **surpasses GPT-5.4 and matches Gemini-3.1-Pro** while delivering about 2,400 tokens, a tenth of what they read.

Business project: LLM post-training for in-cabin AI (sole owner)

- **SFT:** LoRA fine-tuning on Qwen3-4B / 8B and Qwen3.6-35B-A3B (MoE) with ms-swift; implemented a set-prediction loss (minimum-permutation CE), hard-case weighted SFT and rejection-sampling fine-tuning.
- **Preference optimization and RL, with reward design:** DPO and triplet-level DPO (preference pairs split by Hungarian alignment), GRPO / DAPO; designed a hierarchical multi-component GRPO reward (format validity, schema, hierarchical triplet F1, direction, relation class and process rewards).
- **Evaluation:** built a triplet evaluator (Hungarian matching, placeholder equivalence, relation-synonym normalization, per-class and per-step reporting) with batched vLLM inference; audited and repaired the business memory evaluation set, replacing LLM-based pairing with a deterministic memory-level metric.

Research tracking: responsible for following new papers at top venues such as EuroSys and ICML and presenting them internally.

Fondazione Bruno Kessler (FBK)

Mar 2025 - Mar 2026

Research Intern

Trentino-Alto Adige, Italy

- **Proposed Active Instance Verification and built its first benchmark, PlnVerify:** 3,000 episodes over 18 categories in photorealistic 3D scenes, with trap views and unreachable sectors.
- **Evaluation protocol:** positive, same-category and cross-category pairs balanced 1:1:1 to isolate calibration bias; steps-to-decision and navigation-failure metrics; disjoint train and test pools; 95% CIs located the bottleneck in detection.

- **Two baseline tracks:** a training-free modular pipeline, and an end-to-end Qwen3-VL-4B + LoRA agent post-trained on 22K self-built CoT samples with SFT, DPO, GRPO and GSPO, rising from 70.6% to **85.6%**.

Chengdu Jiaoda Guangmang Technology Co., Ltd.

Jul 2022 - Sep 2022

Algorithm Intern

Chengdu, China

- Industrial anomaly detection: GAN-based image reconstruction and object detection for defect inspection on high-speed rail components.

Publications

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"

IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2026 · First author · [IEEE Xplore](#) · [Project Page](#) · [arXiv:2607.03869](#)

Referring segmentation as typed spatial-program execution, no referring supervision: 58.86 / 55.27 mIoU on RRSIS-D / RISBench, over $2\times$ / about $1.7\times$ the best prior training-free results.

Yuhang Jiang, Guohui Deng*, Miaozhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang

"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"

Transactions on Machine Learning Research (TMLR), 2026 · First author · [Project Page](#) · [arXiv:2606.22516](#) · [OpenReview](#)

Resize-based input diversity, a default in transfer attacks, reverses sign on adversarially trained surrogates, so using it blindly overstates robustness.

Yuhang Jiang, Xiaojing Chen

"PlnVerify: An Offline Embodied Benchmark for Active Instance Verification"

Poster, FMEA Workshop @ CVPR 2026 · Sole author · [Project Page](#) · [arXiv:2605.30639](#) · [OpenReview](#)

First benchmark for Active Instance Verification; active viewpoint selection gives no reliable gain, and detection quality binds the task.

Yuhang Jiang · Master's thesis project

Preprints and Other Publications

"Retrieved but Not Delivered: Multimodal Memory Delivery for Long-Term Agents"

[arXiv:2609.32590](#) · [Project page](#) · under review

The bottleneck of multimodal memory is delivery, not retrieval; training-free DeliverMem leads the strongest published methods on MemLens and DMV-Bench.

Yuhang Jiang, Qingwei Liao, Kaize Yin, Xingling Liu, Luca Cuomo, Silvio Bacci

"A Circuit, Not The Circuit: Non-Unique Causal Localisation of the Mamba-2 State Sink"

[arXiv:2606.00930](#)

Names the state sink, the state-space analogue of the attention sink; a near-disjoint set of heads matches the causal effect of the probe-selected ones, so a probe recovers a circuit rather than the circuit.

Yuhang Jiang, Bowen Zhang

"Refit the Probe: Single-Direction Ablation Is Not a Necessity Test"

[arXiv:2606.00926](#)

A probe refitted on the ablated activations recovers its accuracy, so single-direction ablation cannot establish necessity; subspace erasure against matched random subspaces reverses which representations look causally inert.

Yuhang Jiang

"CL-RAG: A Closed-Loop Multimodal Retrieval-Augmented Generation Architecture for Robust Human-Robot Control Interaction"

WRC SARA 2025 (oral) · Bowen Zhang, Yuhang Jiang, Lingxiang Hu, Dun Li, Qianqian Hu*

Note: * indicates the corresponding author.

Awards and Competitions

- **Kaggle Competitions Expert** (all solo): Silver, March Machine Learning Mania 2025 (64 / 1,727); Bronze, BirdCLEF+ 2025 (199 / 2,025); Bronze, BirdCLEF 2024 (70 / 974).
- **Mathematical Contest in Modeling (MCM), Finalist** (top 1%), 2021. Team leader; responsible for modelling and implementation. First winner from the School of Internet and youngest winner in the history of Anhui University.

Skills

- **Research areas:** multimodal LLMs, agent memory, LLM agents, RAG, embodied AI, mechanistic interpretability.
- **LLM post-training:** SFT, DPO, GRPO / GSPO / DAPO, RFT, RL reward design, LoRA / PEFT.
- **Frameworks and systems:** PyTorch, Transformers, vLLM, ms-swift; scikit-learn, MPI / OpenMP (HPC).
- **Programming:** Python, C/C++, Matlab, R.
- **Service and languages:** Reviewer, ICLR 2027; FMEA Workshop @ CVPR 2026. English (fluent, TOEFL 102); Chinese (native).